

Part A: Fundamentals

1. What is Data Analysis?

Data Analysis is the process of collecting, cleaning, and analyzing data to extract useful insights and support decision making. It involves organizing raw data into meaningful information using statistical and computational techniques. In simple terms, it helps us understand patterns, trends, and relationships present in the data. For example, a company can analyze customer data to understand buying behavior and improve their services.

2. Steps in a Data Science Project

A Data Science project follows a systematic approach. First, we define the problem clearly. Then we collect data from different sources such as CSV files, databases, or APIs. Next, we perform data cleaning to handle missing values and remove inconsistencies. After that, we explore the data using visualization techniques (EDA) to understand patterns. Then, we build machine learning models using suitable algorithms. After model building, we evaluate the performance using metrics like accuracy. Finally, we deploy the model and monitor its performance in real-world applications.

3. Machine Learning Problem Statement

The objective of this project is to predict whether a customer will churn based on their purchase behavior and demographic details. The dataset includes features such as age, income, purchase frequency, and average purchase value. Using these features, we build a classification model that categorizes customers into churn (1) or not churn (0). This helps businesses identify customers who are likely to leave and take necessary actions to retain them.

4. What are Tensors?

Tensors are multi-dimensional arrays used to represent data in machine learning and deep learning. They are generalizations of scalars, vectors, and matrices. A scalar is a 0-dimensional tensor, a vector is a 1-dimensional tensor, and a matrix is a 2-dimensional tensor. Tensors can have higher dimensions as well and are widely used in libraries like NumPy and TensorFlow.

In-depth Explanation with NumPy Example

In NumPy, tensors are represented as arrays. For example:

Scalar: `a = 5`

Vector: `a = [1, 2, 3]`

Matrix: `a = [[1, 2], [3, 4]]`

3D Tensor: `a = [[[1,2],[3,4]], [[5,6],[7,8]]]`

These structures allow efficient storage and computation of large datasets. Operations like addition, multiplication, and reshaping can be easily performed on tensors.